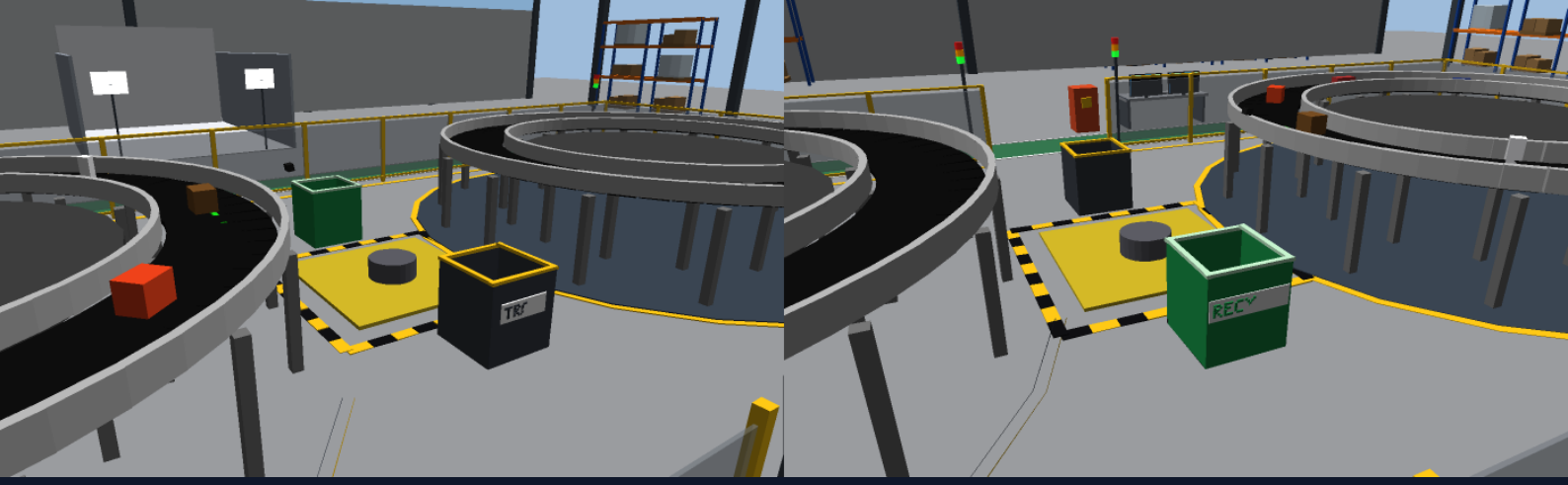



REPORT 04 · FACTORY BENT SIMULATION WITH A BCR ROBOT ARM

# Robot Arm Pick & Place Evolution

Attacher, URDF IK, catch station, close grasp, movement styles and bins

ROS 2 Jazzy · Gazebo Harmonic · OpenCV · PyQt5 · BCR 7-axis arm

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# 1. Where the arm started

Before October 26, the BCR arm was driven from a PyQt5 UI (`bcr_arm_simple_ui`) with preset joint poses. The grasp was faked by `bcr_endpoint_object_attacher`, which moved one model, `pickup_box`, to the gripper with repeated `gz service set_pose` calls. There were three problems. The object sat at a fixed offset (−0.04, 0, −0.15) rather than at the tip. The offset rotated with the gripper. And each update started a new process (50–100 ms), so the box trailed the arm and landed somewhere else on release.

# 2. Exact attach (Oct 26, attacher V2.9)

- `attach_mode = gripper_top`: the object's top face is centred on the `gripper_tip` frame with a 5 mm `contact_gap`, keeping the gripper's yaw.
- Pose updates go through the **gz-transport Python bindings**, with a fallback to the CLI.
- On release the object first freezes at its carried pose, and then it becomes a physics body again.
- `auto_select_target`: the attacher grabs whichever product is closest to the gripper, using that product's own size.

# 3. Inverse kinematics from the URDF

`urdf_ik.py` reads the kinematic chain from `/robot_description`, so it always matches the launched xacro. Forward kinematics composes the joint transforms. Position IK is **damped least squares** on the tip position, with a **null-space pull** toward a taught seed posture (the UI's "Pick Contact" pose), so solutions keep a familiar shape. Later, `frame_positions` was used to reject postures that put any joint, or the middle of any link, within `bin_arm_clearance` (0.15 m) of a bin wall.

# 4. The controller grows (vision_pick_place.py)

| Phase | Behaviour added |
|---|---|
| vision | `mode:=pick_zone`: wait for a still object in a zone, IK approach, attach, lift, place on ring 2 |
| station | hover over the catch station and catch moving products when the ROI tracker predicts entry |
| sidecam | cross-check, then `/hold_object_for_pick` stops the product. Descend to `descend_gap` above its measured top. Recover if a step fails. |
| carry | lift (capped at `lift_max_z` 2.35 m), one swing round to ring 2 (`swing_direction`), settle before release |
| bentstop | stop all of bent 1 while holding. Lift with joint 2 only. Exact place slots on ring 2. |
| ring2 | wait before the zone until the release camera says EMPTY. Stop bent 2, place, start bent 2. |
| home | wait at home (`wait_at`). Six movement styles, with `motion_type` set to auto / cycle / random / a name. |
| grasp | two-step descent and a grip gated by the FK tip distance (≤ 1.5 cm), with a 3 s timeout |
| bins | destination for each type: bent 2, recycling or trash. Swing to the bin and drop. |
| binfuncs / bindrop / binclear | `put_in_trash()`, `drop_in_recycle_bin()`, `deliver()`. Travel at `bin_travel_z` 2.1 m over the bin centre. Clearance check. |
| worlds / exp | worlds without bent 2: only bin products are picked, the rest stay on the belt. Layout per world. |
| lab | subclassed by the vision lab, so a script decides each destination |

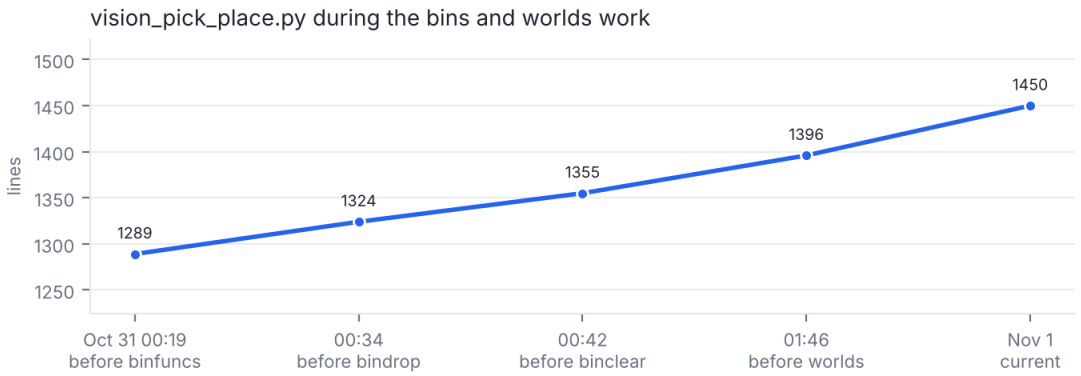



Figure 1. The controller in lines of code across the bins and worlds work on Oct 31.

# 5. Six movement styles

A single fixed movement looked artificial and always used the same side of the cell. `MOTION_TYPES` defines each style by its swing direction, lift, swing time and return path. With `auto`, the style is chosen from the height the side camera measured.

| Style | Direction | Lift | Swing | Return | Used for |
|---|---|---|---|---|---|
| standard | left | 0.50 m | 3.0 s | same way | medium objects (in turn) |
| mirror | right | 0.50 m | 3.0 s | same way | medium objects (in turn) |
| high_carry | left | 0.60 m | 4.0 s | same way | tall (≥ 0.30 m): clears ring 1 |
| low_fast | left | 0.30 m | 2.5 s | same way | flat (≤ 0.15 m): shortest cycle |
| full_turn | left | 0.45 m | 3.0 s | keeps turning left | medium objects (in turn) |
| reverse_return | right | 0.45 m | 3.5 s | left | medium objects (in turn) |

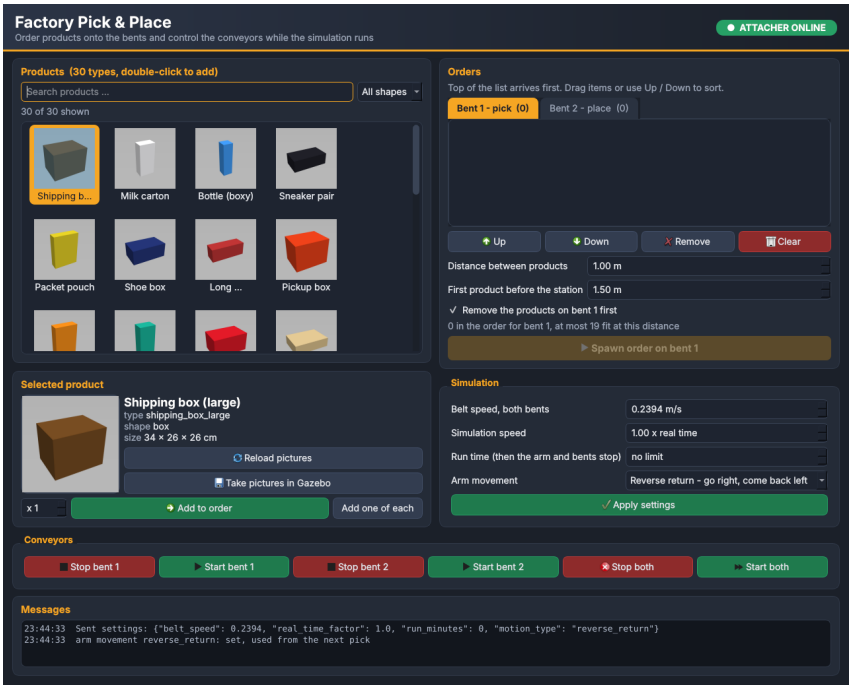



Figure 2. Choosing the arm movement in the order GUI.

# 6. Getting the grasp right

Three problems came up during testing, and each was fixed in its own phase.

| Symptom | Cause | Fix |
|---|---|---|
| The object jumped up to the gripper | The grip fired once every joint was within 0.05 rad. On this long arm that can leave the tip about 10 cm high. | **grasp**: fast to 10 cm above, then slow. Grip only when the FK tip is within 1.5 cm of the contact point. |
| The held product drifted, flickered and turned | Products have no gravity or damping, so the gripper's touch pushed them. | **pin**: re-set the held pose on every timer tick until attached (`pin_held_object`). |
| The product hung over the bin | Gazebo gravity did not take over after release. | **fall**: the attacher moves it down to the bin floor in steps timed like a free fall (≈ 0.5 s). |
| Products hung in the air past the belt end | The belt-end check timed out before Gazebo published poses and crashed the node. | **hangfix**: read the pose stream only, log errors, and retry set_pose. |

# 7. Bins

Two open bins with collisions stand next to the arm: RECYCLE (green) at (0, +2.2) and TRASH (grey) at (0, −2.2). The destination of each product type is saved to `~/.ros/factory_destinations.json` and re-read for every pick. **Auto-sort** sends cardboard, paper, plastic, glass and metal to recycling and the rest to trash. If a bin centre is just out of reach, the tip may stop up to 0.25 m short or go higher, and `release_snap_xy` still drops the product over the centre.

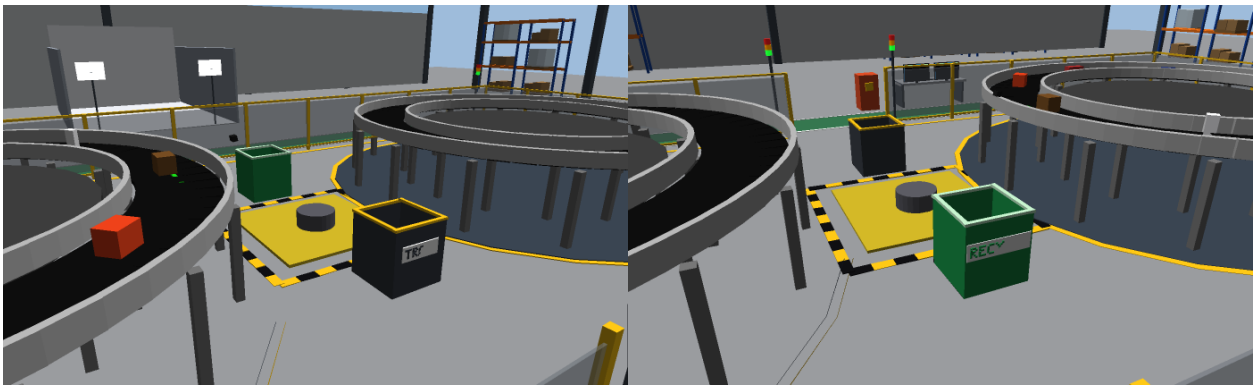



Figure 3. The trash and recycling bins beside the arm.

# 8. Many arms: the pipeline

In the pipeline factory every station uses the same reach (pick and place points 2.45 m from the base, belt top at 1.375 m), and IK was checked at all 12 stations. For the lab version each arm is split into 8 movable link models, posed joint by joint over gz-transport at 20 Hz, so no ROS controllers are needed for 12 arms.




Figure 4. Ring C1 with its arm stations in the pipeline factory.